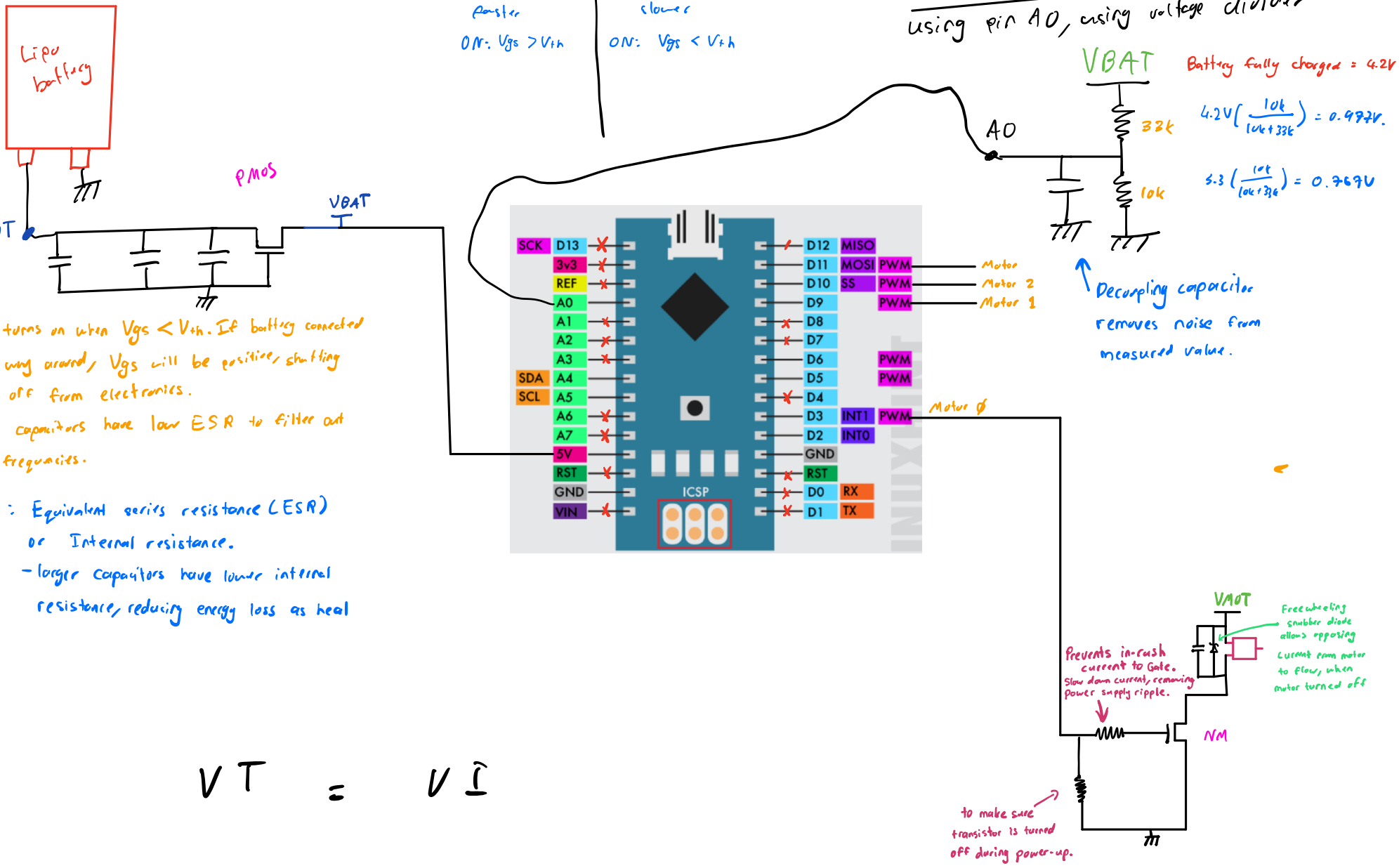
| NMOS | PMOS |
|---|---|
| electron charge faster | hole charge slower |
| ON: $V_{gs} > V_{th}$ | ON: $V_{gs} < V_{th}$ |

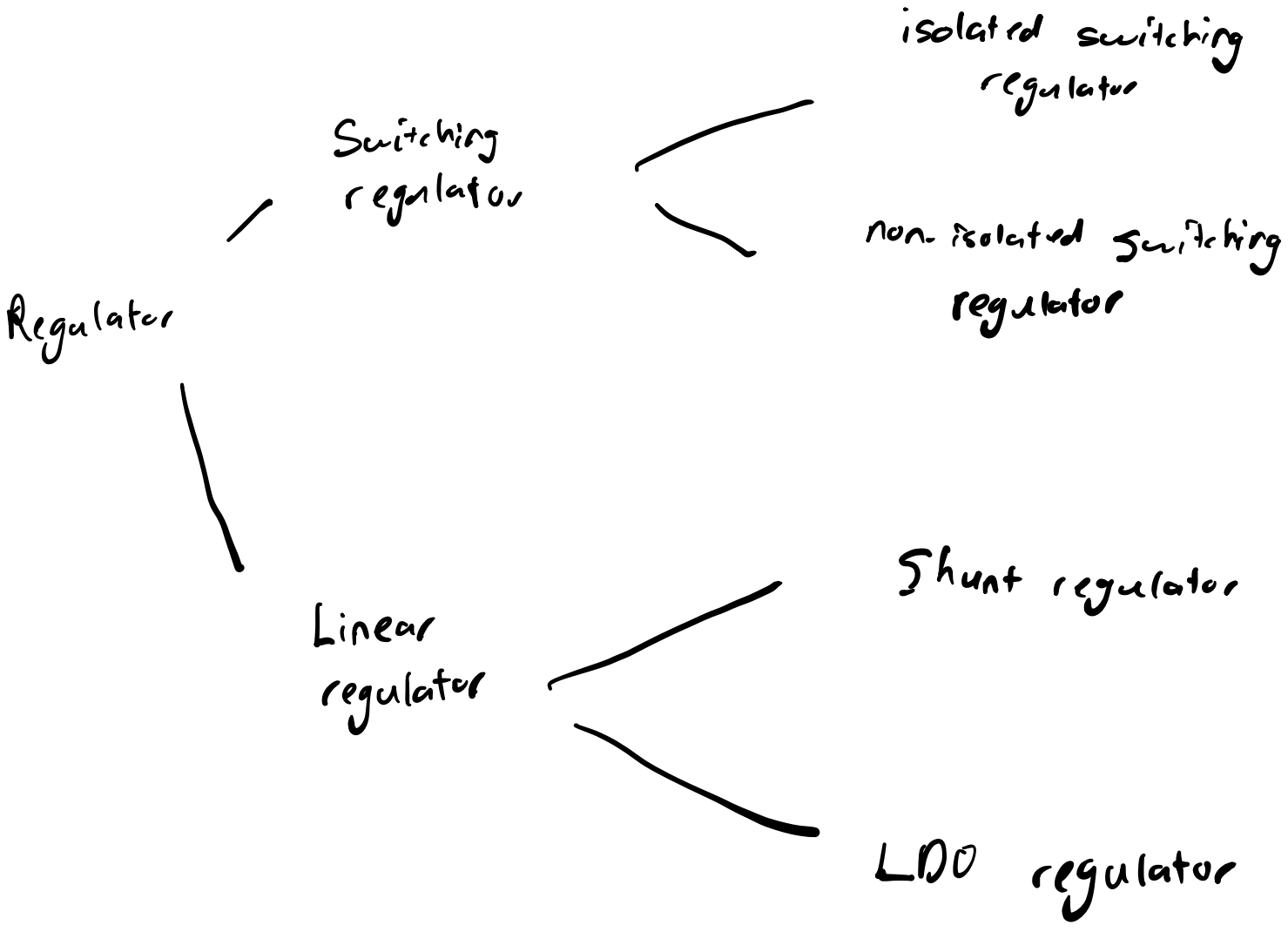
# Battery Detector

using pin A0, using voltage divider




$$V_T = V_I$$

$$= 3.3(7)$$


Regulator
Switching regulator
isolated switching regulator
non-isolated switching regulator
Linear regulator
Shunt regulator
LDO regulator

Using an appropriate board layout is effective in mitigating noise. For instance, optimizing the placement of capacitor and inductor and/or wiring. For more information on the mechanism of how noise (ringing) is generated and how it is controlled, refer to the Application Note "Step-Down Switching Regulator Noise Countermeasures."

### Comparing switching regulator and linear regulator characteristics

| | Switching regulator | Linear regulator |
|---|---|---|
| Output voltage conversion system | Step-down, step-up, step-up/down, invert | Step-down only; $V_{OUT}$ must be less than $V_{IN}$ |
| Efficiency | High (negligible heat generation) | Comparatively low (high heat generation)<br>Low when the difference between input and output voltage is large |
| Output current | Large (high efficiency means high current) | Small |
| Noise | Large | Small |
| Output ripple | Present | None |
| Required external components | Many<br>$C_{IN}$, $C_{OUT}$, L, (SBD) | Few<br>$C_{IN}$, $C_{OUT}$ |

## The ideal switching regulator for any application

Ultra-compact DC-DC converter ideal for automotive cameras

ICs)
› List of switching r (DC-DC converters)
› Step-down switch regulator noise countermeasures (A Note)
› Global Sales Offic
› Product inquiry

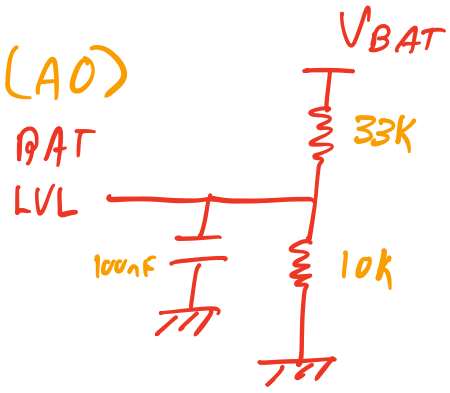
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# Coding:

## Reading Battery voltage

→ Arduino uses power supply as voltage reference.

→ Need to set internal voltage reference. (1.1V)




When Battery fully charged (4.2V).

Potential divider 10k and 33k

$4.2V \left(\frac{10k}{10k+33k}\right) = 0.977V$

Code to read actual battery value and printing to Serial port:

(1) Compensation Factor for physical voltage divider

```
#define VBAT_DIVIDER_COMP (1 / (10k / (10k + 33k)))
```

(2) Internal voltage reference of 1.1V in mV

```
#define BATTERY_VOLTAGE_REF 1100
```

Arduino NaNo ADC is 10-bit → $2^N - 1 = 2^{10} - 1 = 1023$

↳ scales Analog signal range of 0 - 1023

(3) Define Resolution steps, 10-bit gives 1023 resolution steps.

```
#define RESOLUTION_STEPS 1023
```

(4) Equation to get microVolts per LSB

```
#define REAL_BATTERY_MV_PER_LSB (VBAT_DIVIDER_COMP * BATTERY_VOLTAGE_REF / RESOLUTION_STEPS)
```

```
void setup() {                    ← Setup only gets executed once
  setting internal 1.1V reference
  analogReference(INTERNAL)
  Serial.begin(115200);           sets data rate in bps for serial
}                                   transmission.
                                  or
                                  baud rate.

                                  ← loops again and again
void loop() {
  Serial.println(analogRead(BATTERY_PIN * REAL_BATTERY));
}
```

# understanding ADC:

Arduino Nano is 10-bit

↳ has resolution of $2^N - 1 = 2^{10} - 1 = 1023$

This means it can map input voltages between 0 and 5V into integer values between 0 & 1023.

for every $\left[\frac{5V}{1024} = 4.9mV\right]$ per unit

## Flash ADC:

- requires use of cascading high-speed comparators.
  - ↳ N-bit converter circuit uses $2^N - 1$ comparators.
  - ↳ comparing input voltage, to reference voltage.